When Does Graph Learning Beat a No-Arbitrage Constraint?

Coherence and Learnability in Kalshi Event Contracts

Justyn

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Abstract

Mutually exclusive, collectively exhaustive (MECE) baskets of event contracts must be priced to sum to \$1, and threshold ladders must be monotone in strike. These no-arbitrage constraints give free structure that a learned model could exploit. We ask where a spatio-temporal graph attention network (STGAT) adds value over the constraint itself, and where the constraint suffices. Using Kalshi trade data and hourly top-of-book quotes, we find three things. (i) *The baskets are priced coherently.* Where every leg has a two-sided quote, the mid-price sum is within 0.34¢ of \$1. The apparent dislocation in last-trade prices is almost entirely microstructure: 82–87% comes from the side of the book on which trades print, and the rest from one-sided books. (ii) *The constraint is a strong forecaster, and how strong depends on its form.* Shrinking toward \$1 cuts basket forecast error by a third against persistence (shrinkage factor 0.55). The ladder's inequality constraint gives almost no reversion (0.94). (iii) *The STGAT learns traded-price dynamics well, but the skill stops at executable prices.* It forecasts basket trade prices at 0.61× persistence error, and learns settlement convergence without being given time to close; the graph is what lets it compete with the constraint. On the executable bid, however, it is 1.82× the constraint oracle. Its entry timing beats a coin flip but not immediate entry, and no strategy is profitable after fees at quoted prices. The learned model adds value in predicting how prices *trade*; the constraint suffices for what they are *worth*. We also document four evaluation pitfalls that produced false results during the project, and a test that detects each.

1 Introduction

Prediction markets list contracts whose payoffs are tied together by construction. In a MECE basket, such as "Which range will New York's high temperature fall in tomorrow?", exactly one leg pays \$1, so the legs' prices should sum to \$1. In a threshold ladder ("above K_1", "above K_2" with $K_1 < K_2$), the higher strike cannot be worth more than the lower one. These no-arbitrage constraints are known exactly, before any data is seen.

This makes event contracts an unusual test bed for learned forecasting models. In most financial settings, a model must discover cross-asset structure from noisy data. Here the structure is given, and the question becomes sharper:

> *Where does a learned model add value over the constraint itself, and where does the constraint suffice?*

We study this question on Kalshi, a CFTC-regulated exchange, using a spatio-temporal graph attention network (STGAT) whose graph encodes the constraints: basket legs are linked through a basket hub, and ladder strikes are linked in order. We compare it against baselines that use the constraint alone, at two kinds of price: the last traded price, which most studies use, and the executable top-of-book quote, which is what a trader would actually pay or receive.

Contributions.

1. **A coherence measurement.** Using hourly best bid/ask quotes for 10,981 basket legs, we show that Kalshi baskets are priced coherently at the mid. We decompose the apparent last-price dislocation into bid/ask print side and one-sided books. The size of the print-side effect was predicted before it was measured (Section 6.1).
2. **The constraint as a forecaster.** A single shrinkage factor toward \$1 is a strong baseline for baskets and a weak one for ladders. We explain the difference by the constraint's form: an equality pins a level, while an inequality only bounds a half-line (Section 6.2).
3. **Where learning helps.** An ablation ladder runs from persistence, through constraint-only and family-aware shrinkage and a graph-free temporal model, to the full STGAT. It locates the model's value in traded-price dynamics and settlement convergence, and shows that the graph is necessary for it (Section 6.3).
4. **Executable boundaries.** At quoted prices, the model has no forecasting skill on the bid, and its timing beats chance but not immediate entry. No strategy is profitable after fees (Section 6.4).
5. **Evaluation pitfalls.** Four traps, each of which produced a false result during this project, together with the test that detects each (Section 6.5).

2 Background

Prediction markets. Prices of binary event contracts are commonly read as probabilities [13]. They are generally informative, but show systematic biases, most prominently the favourite–longshot bias [11]. On Kalshi specifically, contract prices become more accurate as markets approach close. Takers lose about 32% on average and makers about 10% [1].

Basket dislocations. Saguillo et al. [10] estimate about \$40M of realised profit from MECE baskets on Polymarket priced away from \$1 (April 2024–April 2025). Polymarket was largely fee-free for trading over that period, and the profits went to automated traders acting on the live order book. Whether the same dislocations are executable on a fee-charging, market-maker-intermediated venue is an open empirical question, and one we address.

Microstructure of trade prices. A trade executes at either the bid or the ask, so trade prices "bounce" across the spread even when the efficient price does not move [9]. Trade and quote series can be decomposed into an efficient price plus microstructure noise [2]. Trade direction is classically inferred from quotes [6], with about 85% accuracy [8]. Kalshi instead reports the aggressor side of every trade directly. Constraints spanning several instruments are further affected by non-synchronous trading [7]. Classic put–call parity tests restrict comparisons to near-simultaneous prices for this reason [5, 4].

Graph neural networks for markets. Graph attention [12] and spatio-temporal graph models [14] combine cross-sectional structure with temporal dynamics. In market microstructure, deep models usually forecast *changes* in a reference price, not levels [15]. We use a graph whose edges are the known constraints, not a learned adjacency.

3 Motivation

3.1 From pairs trading to structural constraints

The project began as a statistical pairs-trading study ("Direction A"): search the whole Kalshi universe for pairs of contracts whose prices co-move, estimate the relationship from history, and

trade divergences. A correlation- and half-life-based baseline on October 2025 data returned a Sharpe ratio of 5.52. On inspection, most selected pairs were economically unrelated, for example a music-chart market paired with a public-health case count. The result is best explained by multiple testing. With a 20-day lookback, a correlation of about 0.44 is nominally significant, and tens of thousands of candidate pairs were screened without correction.

Three structural problems make estimated relationships a poor fit for event contracts:

- Most contracts live days or hours, far shorter than any formation window.
- Binary terminal payoffs have no stationary equilibrium spread to revert to.
- Screening millions of tickers guarantees spurious pairs.

On supervisory advice, the scope moved to relationships that need no estimation: the MECE and ladder constraints, which are exact, known in advance and enforced at settlement. A deviation from them therefore has a defined target, and the question of whether it can be forecast is well posed.

3.2 Why the first positive result did not stand

Early in this project, a model trained on last-trade prices appeared to beat an optimal constraint-based shrinkage forecast by 15%. It also appeared to identify tradeable dislocations. Both claims failed under scrutiny, for reasons that motivate the design of this paper.

- **Last-trade basket sums are dominated by microstructure.** 68.7% of basket trade prints are ask-side. With a half-spread of 2.8¢, that predicts a +6.3¢ mean excess on a 6-leg basket. The measured excess was +5.25¢. A model that forecasts last-trade sums may therefore be forecasting which side the next prints land on, not mispricing.
- **Several pipeline leaks inflated the result.**
  - resolved markets were replayed after close;
  - time-to-close used after-the-fact close times;
  - basket membership was selected with hindsight;
  - baskets were scored on a subset of their legs.

  Section 6.5 quantifies each.

We therefore (a) rebuilt the pipeline causally and pre-registered every re-run before seeing results; (b) scored forecasts against baselines that use the constraint, not only against persistence; and (c) measured executable prices directly from quotes rather than inferring them from trades.

4 Method

4.1 Population and graph

Contracts are nodes on a 2-hour snapshot grid. A *MECE basket* is an event whose legs partition the outcome space. Membership is decided causally in validation and test: a family qualifies only on instances known to have resolved at least 24h earlier, and the event's own outcome is never used. (Training months use a curated membership list.) Each basket has a hub node, linked to its legs by directed edges in both directions. *Ladder* pairs are adjacent strikes of the same underlying, linked by a monotonicity edge. Node features summarise each 2-hour window: last price, VWAP, return, volatility, volume, order flow and hub leg counts. Time-to-close is zeroed as an input in every split, and ladder edge features are zeroed.

4.2 Model

The STGAT has four parts:

- a type-specific input projection;
- a two-channel spatial attention layer, whose channels are MECE hub–leg and ladder;
- per-node *causal* temporal self-attention over the snapshot sequence;
- a residual forecast head, $\hat{p}_{t+h} = p_t + \Delta_h$, with the final layer zero-initialised so training starts exactly at persistence.

Horizons are $h \in \{1, 2, 3, 6, 12\}$ snapshots, i.e. 2–24 hours. The loss combines per-node error with a basket-sum term; the ladder term's weight is zero. Checkpoints are selected on validation MECE error.

4.3 Targets and baselines

For basket b with legs L_b, let $S_t = \sum_{i \in L_b} p_{i,t}$. Under the `whole_settle` rule, a basket is scored only if every listed leg is present. A leg that settles by $t + h$ is valued at its settlement (0 or 1). Baselines:

- **Persistence:** $\hat{S}_{t+h} = S_t$.
- **Constraint oracle:** $\hat{S}_{t+h} = 1 + \alpha(S_t - 1)$, with the single α that minimises squared error *on the scored split*. This is deliberately optimistic.
- **Family oracle:** the same, with one α per series $\times$ horizon. A basket's family is known in advance.

4.4 Executable prices

Hourly best bid and ask for every basket leg come from Kalshi's public historical candlestick endpoint [3], aligned to the snapshot window ends. A bid of 0, or an ask of \$1, is a real one-sided book and is kept as the executable price.

- **Bid-side forecast.** The model's predicted change in the last-trade sum is applied to the quoted bid sum: $\hat{B}_{t+h} = B_t + (\hat{S}_{t+h} - S_t)$. The half-spread cancels in the change if the spread is stable [9].
- **Hold to settlement.** Selling a basket at the bids yields gross $(\sum_i \text{bid}_i - 1)$ per set, and buying at the asks yields $(1 - \sum_i \text{ask}_i)$. Kalshi's taker fee is charged on the executed price.

4.5 Timing control

The model chooses to enter now, or to wait one step if it predicts the dislocation will widen. The control makes the same choice by coin flip, on identical events and sizes, so any gap is due only to the forecast. Confidence intervals come from a cluster bootstrap over baskets.

4.6 The causal rebuild and how each fix was verified

The first pipeline leaked future information in five ways. Each fix was checked with a test that has a known answer, not assumed to work (Table 1).

Residual risks we are aware of.

- Training months use a curated membership list. This affects which baskets the model learns from, not evaluation.

Table 1: Leakage fixes and their verification.

Leak	Fix	Verification
Resolved markets replayed after close	drop trades after close; no carry-forward past close	past-close share of scored ladder pairs falls from 66% to 0%
Time-to-close from after-the-fact close times	input slot zeroed in every split	randomising the slot changes predictions by exactly 0
Check: can temporal attention see the future?	causal mask (already present)	scrambling every snapshot after t_0 leaves predictions at $t \leq t_0$ bit-identical
Basket membership chosen with hindsight	causal membership in validation/test	event's own outcome never used; family must have resolved ≥24h earlier
Partial baskets scored as whole	score only baskets with every listed leg (`whole_settle`)	re-score reproduces the scorer chunk by chunk (reproduction guard)
Ladder edge features from whole-life statistics	zeroed	—

- The `--min-ttc` robustness filter reads recorded close times, which include early closes. Results using it are labelled ex post.
- The unsettled stratum conditions on a basket not settling within h, which is not strictly known at t (scheduled close times are public, early closes are not).
- September 2025 has been scored for several model variants. Each protocol was pre-registered and scored once, but the multiple-look caveat remains until the walk-forward supplies fresh test months.

5 Experimental Setup

- **Data.** Kalshi trades and markets, May–September 2025, with a strictly forward split: train May–June, validation July–August, test September. Hourly quotes cover all 10,982 validation/test basket legs; 10,981 were retrieved.
- **Seeds and reporting.** Four seeds per configuration. We report the mean and range over seeds, with a 95% cluster-bootstrap CI per seed and the cluster count k.
- **Pre-registration.** Every re-run reported here was pre-registered in the repository before results were seen: the protocol, the decision rule and a stated prediction. The test split was scored once per seed per protocol. Deviations are disclosed.
- **Test positions.** 37,392 basket-horizon positions ($k = 348$ baskets), of which 18.6% settle within the horizon.

6 Results

6.1 Baskets are coherent at quoted prices

Where every leg has a two-sided quote, the mid-price sum is within 0.34¢ of \$1 (Table 2). The same holds on validation (+0.50¢). In these baskets, 82–87% of the last-trade excess is print side. Across all baskets, the remaining excess sits in legs with no bid, whose mid is inflated. Executable dislocation is rare and small. Selling at the bids beats \$1 in 1.8% of snapshots, by about 2¢, before a taker fee of 1.3–5.8¢ per basket. The quoted per-leg spread (median 2.0¢, mean 2.9¢) is below our earlier trade-based estimates (4.7–5.6¢). The trade-print bid proxy overstated quoted bids by 1.1¢ per leg on average.

Table 2: Basket sums on the test split: whole baskets with every leg live. Quotes cover 98.1% of leg-snapshots and 97.1% of basket-snapshots.

	all baskets ($n = 7{,}532$)	every leg two-sided (69%)
Sum of last trades	1.0405	1.0261
Sum of quoted mids	1.0185	**1.0034**
Sum of bids / asks	0.944 / 1.093	0.967 / 1.039
Sell at bids beats \$1	1.8% of snapshots ($\approx 2¢$)	—

6.2 The constraint as a forecaster

On test, the constraint oracle has 0.67× the error of persistence for basket sums ($\alpha = 0.55$), but 0.985× for ladder gaps ($\alpha = 0.94$). Basket deviations revert strongly to the level the constraint pins. Ladder gaps are only bounded above by zero, so there is no level for them to revert to.

6.3 Where learning helps: the ablation ladder

Table 3: Test error, relative to the single-α constraint oracle unless stated. Mean over 4 seeds (range). "Unsettled" = baskets with no leg settling by $t + h$.

Model	all positions	unsettled	vs. persistence
Persistence	1.49	—	1.00
Constraint oracle (single α)	1.00	1.00	0.67
Temporal model, no graph	1.128	1.106	0.760
STGAT (full)	**0.909** (0.883–0.941)	1.034	**0.613**
STGAT, unsettled-only retrain[a]	—	0.950	0.781
Against the family oracle (per series × horizon), unsettled:			
STGAT, unsettled-only retrain	—	1.052 (1.024–1.074)	—
Temporal model, no graph	—	1.224	—

[a] Settled baskets excluded from the loss and from selection.

Three findings (Table 3):

- **The STGAT forecasts traded basket prices well**, at 0.61× persistence error, consistently across seeds.
- **The graph is necessary.** Without the MECE and ladder edges, the model falls behind even the single-α constraint (1.128×).
- **On test, the margin over the constraint comes from settlement convergence.** Baskets that settle by $t + h$ have a target of exactly \$1, and the model anticipates this even though time-to-close is not an input. Perturbation tests confirm that predictions at t are unchanged when future snapshots or time-to-close are randomised, so this is not leakage. The model's pull toward \$1 rises smoothly as close approaches, with no jump at the settlement boundary, which points to family-level lifecycle learning. On unsettled baskets the model matches, but does not beat, the constraint plus contract family. On validation it does beat it (0.887× single-α unsettled), but that is the selection split.

Ladders (exploratory). On 99,646 test ladder positions, the model scores 0.982–0.993× the oracle and 0.966–0.977× persistence. The graph-free model is marginally better (0.974–0.982×), so the small ladder gain comes from per-contract temporal forecasting, not from ladder structure.

6.4 Executable boundaries

Table 4: Executable-price results on test (4 seeds, cluster CIs by basket).

Quantity	result	CIs excluding null
Bid-side forecast / oracle, unsettled[a]	1.822× (1.80–1.85)	4/4 above 1
Timing: model − random, last prices[b]	+0.192 (0.179–0.211)	4/4 above 0
Timing: model − random, quoted prices	+0.867	3/4 above 0
Timing: model − enter now, quoted prices	−0.957	4/4 *below* 0
Fee-aware entry at quotes: trades fired / net	1 of 2,090 / +\$0.07	—
Static entry at quotes: net per trade	−\$2.18	—

[a] $n = 30{,}001$ positions, $k = 344$ baskets. [b] \$/trade, $k = 312$ baskets; same units in the rows below.

Three results (Table 4):

- **No bid-side skill.** Applied to the quoted bid, the model's forecast is 1.82× the constraint oracle, and worse than persistence. Its predicted change tracks where trades print, not where the bid moves.
- **Timing beats chance, but not the trivial policy.** The model's entry timing beats a coin flip at last prices (4/4 seeds) and at quoted prices (3/4). At quoted prices, however, waiting is costly, and the model loses to entering immediately on every seed.
- **No profit after fees at executable prices.** A rule that enters only when the quoted edge exceeds the fee fired once in 2,090 opportunities. This agrees with evidence that Kalshi takers lose on average [1]. It contrasts with the fee-free Polymarket evidence [10].

6.5 Evaluation pitfalls

Table 5: Traps that produced false results during the project, and the test that detects each.

Trap	Effect	Detection
Settled contracts replayed after close	57% of old scored ladder pairs (train/validation months) were frozen replays, producing "ladders are unforecastable"	audit of the share of pairs past close
Settlement-pinned targets	all of the 0.909× test margin came from baskets settling within h	stratify by settlement
Last trades treated as executable	bids overstated by 1.1¢ per leg; spreads by ≈40%; bid-side skill assumed	validate against quoted top of book
Ablation scored with the wrong graph	graph-free checkpoints would have been scored *with* the graph	record dropped edges; known-answer test

These were caught by reproduction guards, which check that every re-score reproduces the committed figures exactly, and by pre-registration (Table 5). One headline figure had no log at all, and was recovered and found to have been computed on the selection split. The provenance of every reported number is now logged.

7 Discussion and Limitations

- **Answer to the question.** Graph learning adds value on traded-price dynamics: where trades print, and how baskets converge toward settlement. The graph is necessary for this.

Wherever money changes hands, the constraint suffices. This follows from the coherence finding: if baskets are coherent at executable prices, no model should profit there.

- **Limitations.**
  - Top-of-book quotes only, at hourly resolution; no depth, so fills at size are assumed.
  - A single test month, scored for several model variants; the multiple-look caveat applies.
  - The timing control's wait rate is not matched to the model's.
  - Ladder figures are exploratory.
  - The pilot window is five months, and the population is concentrated in weather and sports.
- **Next.** A pre-registered 7-fold walk-forward over November 2024 to November 2025, with the protocol frozen from this study, fresh held-out months, and quotes for all months.

8 Questions for Supervisor Review

8.1 Update on the Recess Week consultation (24 September)

The consultation document raised seven questions. Its three supporting results ($0.853\times$ the oracle; about 92% of the margin from the graph; directional skill of +\$2.92 per trade over random) all came from the leaky pipeline, and are superseded by the results above. Table 6 gives the current status of each question.

8.2 Data-window memo (26 September): still applicable

All six questions remain open and determine the walk-forward design:

1. Does a seven-fold expanding walk-forward within one liquidity regime (November 2024–November 2025) address the concern, or is the concern calendar span?
2. Is 2022-04 to 2022-11 acceptable as a labelled structural backcast?
3. Should the thin 2022-12 to 2024-10 period be included with a staleness filter, or excluded?
4. Is the basket-day the right unit for reporting effective sample size?
5. Is there a departmental expectation on window length?
6. *Revised.* The proposed dissertation shape has changed. It is now (a) the coherence and microstructure measurement, strengthened with quotes; (b) the constraint as a forecaster, plus the ablation ladder (the "92% from the graph" attribution is retired); and (c) the ladder as a contrast case corrected for an artefact, not a diagnosed negative control. See concern A.

8.3 New questions

A. Is the reframing valid? The proposed research question is: *where does a learned model add value over the constraint, and where does the constraint suffice?* Its positive answers are:

- coherence at quoted mids;
- the constraint as a strong forecaster whose strength depends on its form;
- the graph being necessary for the model's traded-price skill;
- timing better than chance.

Table 6: Recess Week consultation questions: what has changed since.

Question (then)	What has changed	Status
Q1. Does the evidence meet the requirement to beat baselines *and show PnL*?	It beats persistence (0.61×) and matches constraint-plus-family shrinkage. Its timing beats a coin flip on held-out test (+\$0.19 per trade, 4/4 seeds). No strategy is profitable after fees at quoted prices. The PnL half of the requirement is not met.	**Open**; see new concern C below
Q2. Are the execution proxies (spread, depth, maker fills) adequate?	Partly resolved. Kalshi's historical API *does* retain hourly top-of-book quotes; the consultation stated otherwise. Against real quotes, the trade-based spreads were ≈40% too high and the trade-print bid proxy 1.1¢ per leg too high. *Depth* is still unobservable, so the participation cap remains.	Open, narrowed to: is top-of-book at hourly resolution adequate?
Q3. Keep static entry as a baseline?	At quoted prices, static entry is now *executable* at top of book, not idealised. On identical events it beats the model's timing by \$0.96 per trade (4/4 seeds). This is the matched-event comparison Q3 proposed.	Proposed: keep it as a full baseline
Q4. Is gross edge a legitimate measure of signal quality?	Gross edge at last prices was |dev|-based, and so mostly overround. At quotes it is negative on 97% of trades. We propose measuring signal quality by forecast error ratios and paired timing gaps instead, with net PnL reported alongside.	Proposed: replace gross edge
Q5. Add features to predict *whether* a ladder pair reprices?	Time since last trade predicts repricing (AUC 0.83) but is not a model input. Not pursued, to protect the deadline.	Open (future work)
Q6–7. Abandon the ladder? Still document it?	The ladder negative was an artefact: 57% of the old scored ladder pairs (train/validation months) were replayed dead contracts. After the fix, the ladder is slightly positive (exploratory), and the graph adds nothing to it.	See new concern B below

Its negatives are stated as boundaries: no skill at executable prices, and no profit. Is this an acceptable dissertation, or does it read as retreating from the original trading question?

B. Should the ladder stay? Our proposal is to keep it as the contrast chapter. An inequality constraint gives weak reversion ($\alpha = 0.94$ against 0.55), the graph adds nothing, and the earlier "unforecastable" verdict was a replay artefact, itself a methodological finding. Its figures are exploratory and would be firmed up in the walk-forward. Alternatively, we could drop it to a short section for a focused MECE study. Which do you prefer?

C. Is positive PnL a hard requirement? After fees, at quoted prices, no strategy is profitable. The fee-aware rule fired once in 2,090 opportunities. This agrees with Kalshi-wide evidence that takers lose about 32% on average [1]. We do not believe a profitable strategy exists in this data without order-book speed and depth that we do not have. If positive PnL is required, the project cannot meet it honestly. If rigorous evidence on *why* it is absent is acceptable, Sections 6.1 and 6.4 provide it.

D. Are the leakage fixes sufficient? Section 4.6 lists each fix, its verification and the residual risks. Is there any check you would want before the walk-forward is run?

E. Walk-forward scope. Seven folds need about 21 hours of training, and quotes for an estimated 30–40k legs. Should we pre-register all seven, or a smaller fixed subset (e.g. 3–4

folds) to protect writing time?

9 Conclusion

Kalshi's MECE baskets are priced coherently once microstructure is accounted for. A no-arbitrage constraint is a strong forecaster where it pins a level, and a weak one where it only bounds an inequality. A constraint-structured STGAT learns traded-price dynamics well, and needs its graph to do so. That skill does not extend to executable prices, where the constraint alone is as good or better.

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